

RecycleVision

Garbage image classification using deep learning: from a CNN built from scratch to a fine-tuned ResNet50 deployed as a live web app.

97.4%

Test accuracy (ResNet50)

0.968

Macro F1 across 12 classes

15,515

Training images

3

Models compared

battery



biological



brown-glass



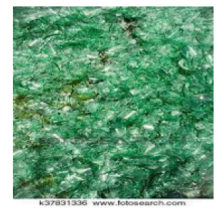
cardboard



clothes



green-glass



metal



paper



plastic



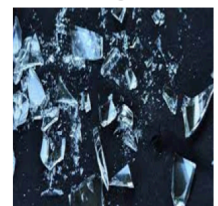
shoes



trash



white-glass



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Live app recyclevision-garbage-image-classification-using-deep-learning.streamlit.app

Source code github.com/mukundnamesh2605/RecycleVision--Garbage-Image-Classification-Using-Deep-Learning

Stack Python 3.11 · PyTorch · torchvision · scikit-learn · Streamlit

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1 Executive summary

RecycleVision classifies a photo of a single waste item into one of **12 categories**: battery, biological, brown glass, cardboard, clothes, green glass, metal, paper, plastic, shoes, trash and white glass. The goal is to automate the first step of recycling, which is working out what an item is so it can go to the right stream.

Three models were trained and compared on an identical, stratified split of 15,515 images:

MODEL	TEST ACCURACY	MACRO F1	PARAMETERS
Baseline CNN (from scratch)	78.05%	0.735	~0.62 M
MobileNetV2 (transfer learning)	95.96%	0.945	~2.2 M
ResNet50 (transfer learning) ✓ deployed	97.38%	0.968	~23.5 M

Key takeaways

- **Transfer learning is decisive.** ImageNet-pretrained backbones added about 18–19 accuracy points over a CNN trained from scratch.
- **Two-phase training pays off.** Training only the new head first, then fine-tuning the whole network at a 10× lower learning rate, added about 2 points for both transfer models.
- **Class imbalance is handled.** With a class-weighted loss, even the smallest classes reach F1 above 0.90.
- **Real-world accuracy is lower.** On 120 web-scraped photos, accuracy drops to **77.5%**. The model is strong on clear single items and weaker on cluttered scenes.
- **It's live.** The best model runs in a Streamlit web app that anyone can try in a browser.

2 Problem statement & business use cases

Manual waste sorting is slow, costly and error-prone, and contamination (the wrong material in a recycling stream) lowers the value of everything collected. The aim of this project is to **build a deep learning model that classifies images of waste into material categories**, and to make it usable through a simple interface. It is a building block for automated sorting.

USE CASE	HOW RECYCLEVISION HELPS
Smart recycling bins	A camera in the bin lid identifies the item and opens the matching compartment.
Municipal waste management	Pre-sorting on conveyor lines cuts manual labour and sorting time at recovery facilities.
Educational tools	Instant visual feedback teaches households and schools to separate waste correctly.
Environmental analytics	Logging classifications over time shows waste composition and recycling trends.

Project objectives

1. Prepare and explore a multi-class garbage image dataset, including its class imbalance.
2. Build a preprocessing and augmentation pipeline that every model shares.
3. Train a CNN from scratch as a reference point, then apply transfer learning (MobileNetV2, ResNet50).
4. Evaluate with accuracy, precision, recall, F1 and confusion matrices, and select the model with the best macro F1.
5. Check robustness on images from outside the dataset.
6. Deploy the chosen model as a Streamlit web application.

Scope

Each input is a single image, and the output is one of 12 classes with a confidence score. Locating several items in one image (object detection) is out of scope and discussed in Section 10.

3 Dataset & exploratory analysis

Source: [Garbage Classification \(12 classes\)](#) on Kaggle, with **15,515 RGB images** in one folder per class.

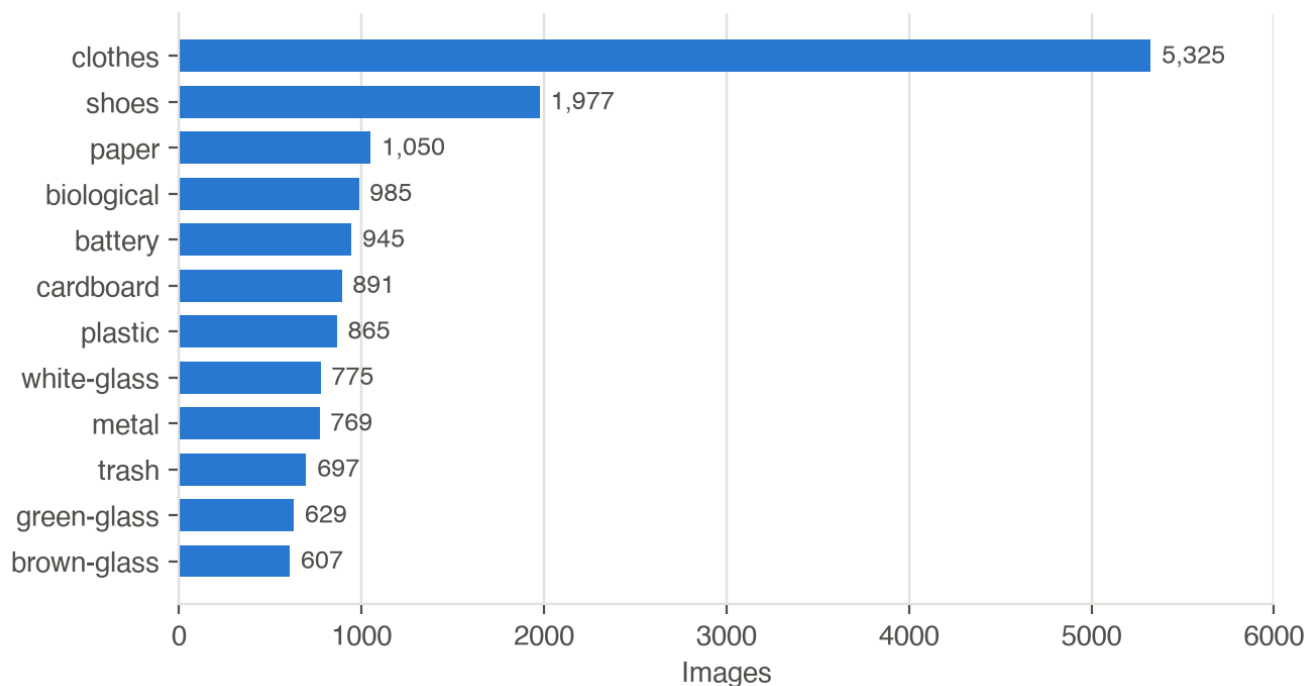


Figure 1. Images per class. `clothes` (5,325) has about 8.8× as many images as `brown-glass` (607).

Finding 1: strong class imbalance

`clothes` makes up 34% of the data and `shoes` another 13%. Trained naively, a model would favour these classes. This is handled with a class-weighted loss (Section 4).

Finding 2: variable image sizes

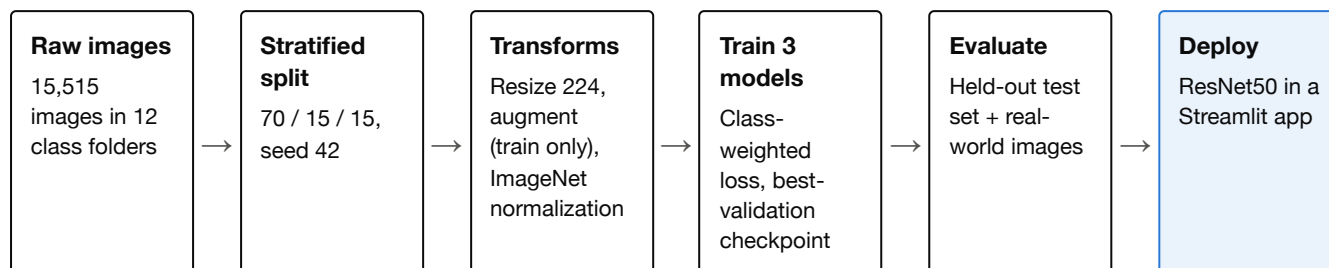
In a random sample of 200 images, widths ranged from 183 to 711 px and heights from 151 to 711 px, with 62 distinct sizes. Every image is therefore resized to a fixed 224×224 input.

Finding 3: visually similar classes

Three glass classes that differ mainly in colour, and several transparent or reflective materials (plastic, white glass, metal), are likely to be confused. The per-class results in Section 6 confirm this.

4 Methodology: data pipeline

All notebooks load data through one shared function, `get_data_loaders()`, so **every model trains and is tested on exactly the same images**. That makes the comparison fair.



Train / validation / test split

SPLIT	SHARE	IMAGES	BATCHES (32)	USED FOR
Train	70%	~10,859	340	Updating weights (augmented)
Validation	15%	~2,328	73	Picking the best epoch
Test	15%	2,328	73	Final, untouched evaluation

The split is stratified, so every class appears in the same proportion in each split. The test set is carved out first, then validation is taken from the remainder.

Preprocessing & augmentation

STEP	TRAIN	VAL / TEST	WHY
Resize to 224×224	✓	✓	Fixed input size expected by the CNN backbones
Random horizontal flip	✓	—	Orientation shouldn't change the class
Random rotation $\pm 15^\circ$	✓	—	Robustness to tilted photos
Colour jitter (brightness, contrast ± 0.2)	✓	—	Robustness to lighting
ImageNet mean/std normalization	✓	✓	Matches the statistics the pretrained backbones expect

Handling class imbalance

Class weights are computed with scikit-learn's `compute_class_weight("balanced")` from the **training labels only** (so no information leaks from the test set) and passed to `CrossEntropyLoss`. A mistake on a rare class therefore costs more:

CLASS	WEIGHT	CLASS	WEIGHT	CLASS	WEIGHT
brown-glass	2.13	metal	1.68	biological	1.31
green-glass	2.05	white-glass	1.67	paper	1.23
trash	1.85	plastic	1.50	shoes	0.65
cardboard	1.45	battery	1.37	clothes	0.24

5 Models & training strategy

MODEL	ARCHITECTURE	ROLE
Baseline CNN	3 × [Conv 3×3 → ReLU → MaxPool] with 32→64→128 filters, adaptive average pooling to 4×4, then Dense 256 → Dropout 0.3 → Dense 12	Reference point: what a model learns from this data alone
MobileNetV2	ImageNet-pretrained (IMAGENET1K_V1); final classifier replaced with Linear 1280 → 12	Lightweight model suited to edge or mobile devices
ResNet50	ImageNet-pretrained (IMAGENET1K_V2); fully connected head replaced with Linear 2048 → 12	High-capacity model aimed at top accuracy

Two-phase transfer learning

Phase 1: feature extraction (10 epochs, lr 1e-3)

The pretrained backbone is frozen and only the new 12-class head is trained. The randomly initialised head learns quickly without damaging the pretrained features.

Phase 2: fine-tuning (5 epochs, lr 1e-4)

The whole network is unfrozen and trained with a 10× lower learning rate, so the backbone adapts to garbage images without forgetting what it learned from ImageNet.

Shared training setup

Adam optimizer · class-weighted cross-entropy · batch size 32 · trained on an Apple Silicon GPU (MPS). After every epoch, the weights with the **best validation accuracy** are kept, so the final model is the best checkpoint rather than the last one. The baseline was trained at lr 1e-3 for 2 warm-up epochs plus 15 epochs.

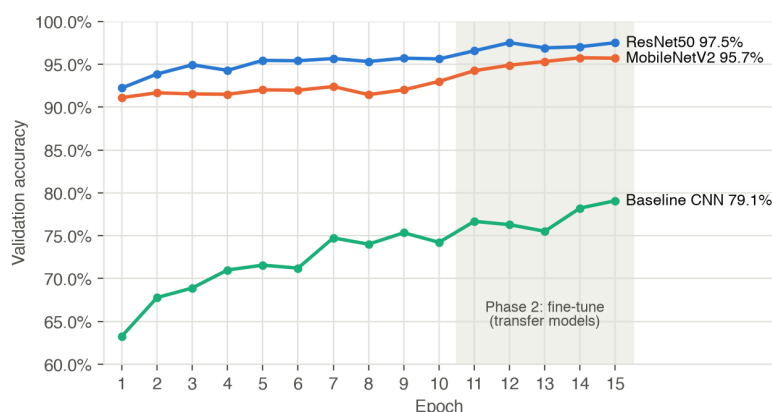


Figure 2. Validation accuracy per epoch. The shaded region is phase 2 (full fine-tuning) for the transfer models. Both transfer models start above 91% after one epoch, while the baseline is still climbing after 15.

BEST VAL. ACC.	PHASE 1	PHASE 2
Baseline CNN	79.1%	—
MobileNetV2	93.0%	95.7%
ResNet50	95.7%	97.5%

Fine-tuning added +2.7 points for MobileNetV2 and +1.8 for ResNet50. The baseline has a single training phase.

6 Results on the test set

All three saved models were reloaded and scored on the same **2,328 held-out test images**, which no model saw during training or checkpoint selection.

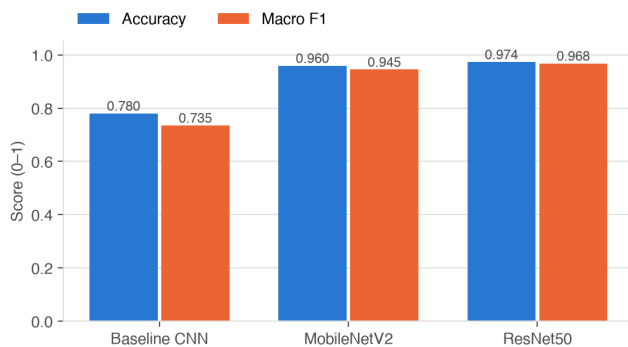


Figure 3. Test accuracy and macro F1 by model.

MODEL	ACC.	MACRO F1	WTD. F1
Baseline CNN	0.781	0.735	0.780
MobileNetV2	0.960	0.945	0.960
ResNet50	0.974	0.968	0.974

Why ResNet50 was selected

It has the highest **macro F1**, which weights all 12 classes equally and is the fairest metric on imbalanced data. MobileNetV2 comes within 1.4 accuracy points at about a tenth of the size, so it is the better choice for on-device deployment.

ResNet50: per-class performance

CLASS	PREC.	RECALL	F1	N
battery	0.947	1.000	0.973	142
biological	0.993	0.986	0.990	148
brown-glass	0.978	0.989	0.984	91
cardboard	0.971	0.985	0.978	134
clothes	0.996	0.980	0.988	799
green-glass	1.000	1.000	1.000	94
metal	0.932	0.957	0.944	115
paper	0.987	0.975	0.981	157
plastic	0.914	0.900	0.907	130
shoes	0.954	0.983	0.968	297
trash	1.000	0.971	0.986	105
white-glass	0.930	0.914	0.922	116
Macro avg	0.967	0.970	0.968	2,328

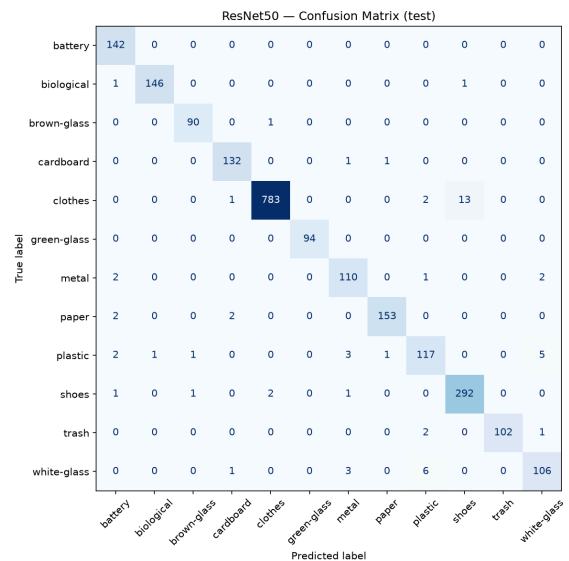


Figure 4. ResNet50 confusion matrix on the test set. Nearly all predictions fall on the diagonal.

Strongest: green-glass is perfect, and biological, clothes and trash are all at or above 0.986. **Weakest:** plastic (0.907), white-glass (0.922) and metal (0.944). These materials are transparent or reflective, and clear plastic bottles in particular resemble white glass. Class weighting worked: the rare classes score about as well as the dominant `clothes` class.

7 Real-world evaluation

Kaggle images are clean, well lit and usually show one object. To test how the model holds up outside that setting, 10 images per class (**120 in total**) were scraped from Bing image search using descriptive queries such as "rotten fruit vegetable" for biological and "metal can" for metal. The deployed ResNet50 was then scored on them with the same preprocessing as the test set.

METRIC	KAGGLE TEST SET	REAL-WORLD IMAGES
Accuracy	97.38%	77.50%
Macro F1	0.968	0.738
Images	2,328	120

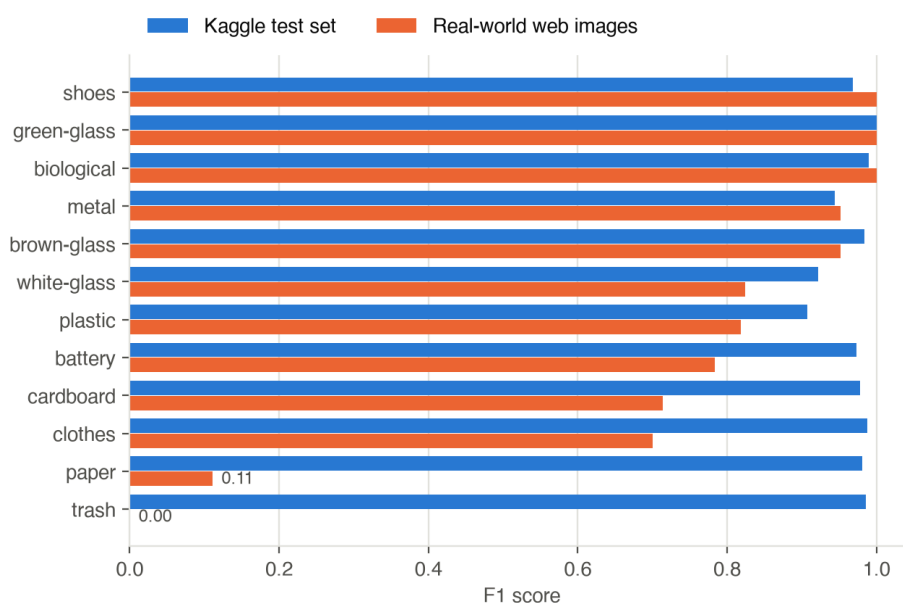


Figure 5. Per-class F1 on the Kaggle test set compared with real-world web images, sorted by real-world F1.

What transfers well

Objects with a distinctive shape or texture keep their performance: **biological, green-glass and shoes stay at 1.000**, and metal and brown-glass stay above 0.95.

What breaks

trash (0.00) and **paper (0.11)** collapse. The "trash" query returned whole scenes of bins and landfills, which look nothing like the single-item Kaggle trash class. **cardboard** becomes a catch-all wrong answer, and its precision falls to 0.556.

How to read this result

The drop has two causes: a genuine **domain gap** (training on clean single objects, testing on cluttered scenes) and a **data-collection artifact** (web queries that don't match the class definitions). The scraped labels were not manually verified, and 10 images per class is a small sample, so treat 77.5% as indicative rather than precise. The model is strong on clear single items and needs more work before it can handle mixed, cluttered waste.

8 The RecycleVision web app

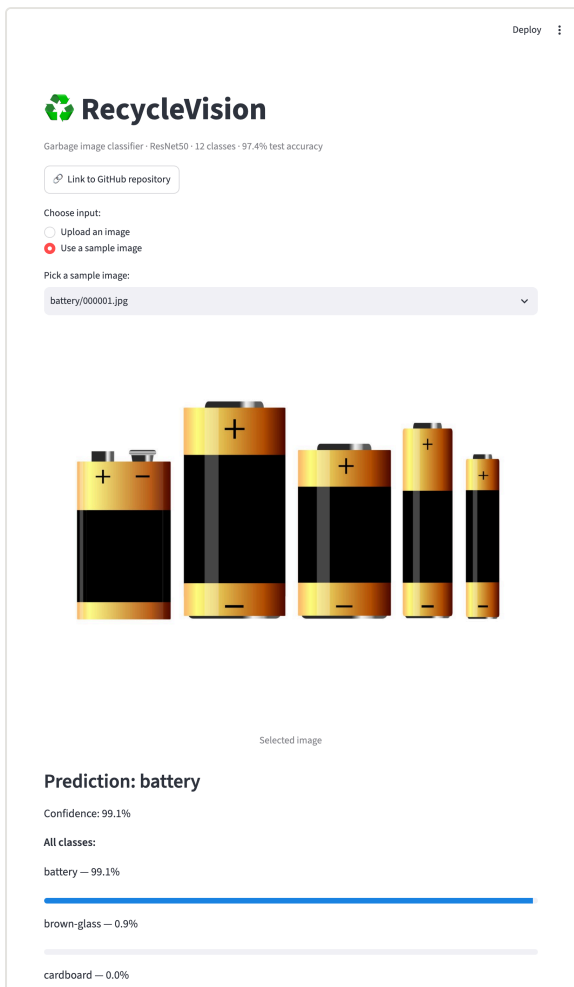


Figure 6. The app in sample-image mode, classifying a battery at 99.1% confidence. Captured from the running app.

Live at recyclevision-garbage-image-classification-using-deep-learning.streamlit.app, hosted on Streamlit Community Cloud.

Features

- **Two input modes:** upload your own JPG or PNG, or choose one of 120 bundled sample images from a dropdown, so no image of your own is needed.
- **Prediction with confidence:** the top class and its probability.
- **Full probability breakdown:** all 12 classes, ranked, each with a progress bar.
- **GitHub link button** in the header.

How it works

1. The image is converted to RGB and resized to 224×224.
2. It is normalized with ImageNet mean and standard deviation, the same as at evaluation time.
3. ResNet50 produces 12 logits, and softmax turns them into probabilities.

Engineering notes

- The model is loaded once and cached with `@st.cache_resource`.
- Inference runs on the **CPU**, so no GPU is needed.
- `requirements.txt` installs the CPU-only PyTorch build, which keeps the cloud deployment small.
- If the sample folder is empty, the app shows a warning instead of failing.

9 Sample predictions

The app's *Use a sample image* mode offers 122 web images from `sample_images/` (12 classes). To show how the deployed ResNet50 behaves on them, **every one of these images** was run through the model with the same preprocessing as `app.py`. The examples below are therefore not hand-picked.

101/122

Sample images correct
(82.8%)

6

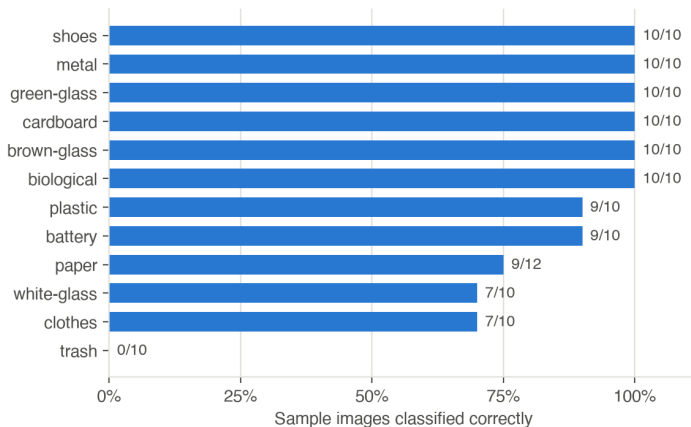
Classes with every sample
correct

8/12

Classes at 90% or better

0/10

Trash samples correct



Six classes are perfect on the samples (biological, brown-glass, cardboard, green-glass, metal, shoes). The others miss 1 of 10 (battery), 3 of 10 (clothes), 3 of 12 (paper), 1 of 10 (plastic), 3 of 10 (white-glass), apart from trash.

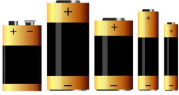
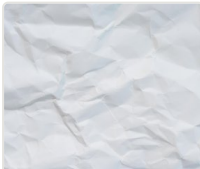
Trash is never recognised. None of the trash samples shows a single item: six are infographics of recycling bins and four are photos of landfills or bin bags.

Paper has been refreshed. The paper folder now holds 12 new images, and 9 are classified correctly. Section 7 was scored on the earlier paper images (F1 0.11), so its paper figures are out of date.

Figure 7. Share of sample images the model classifies correctly, by class.

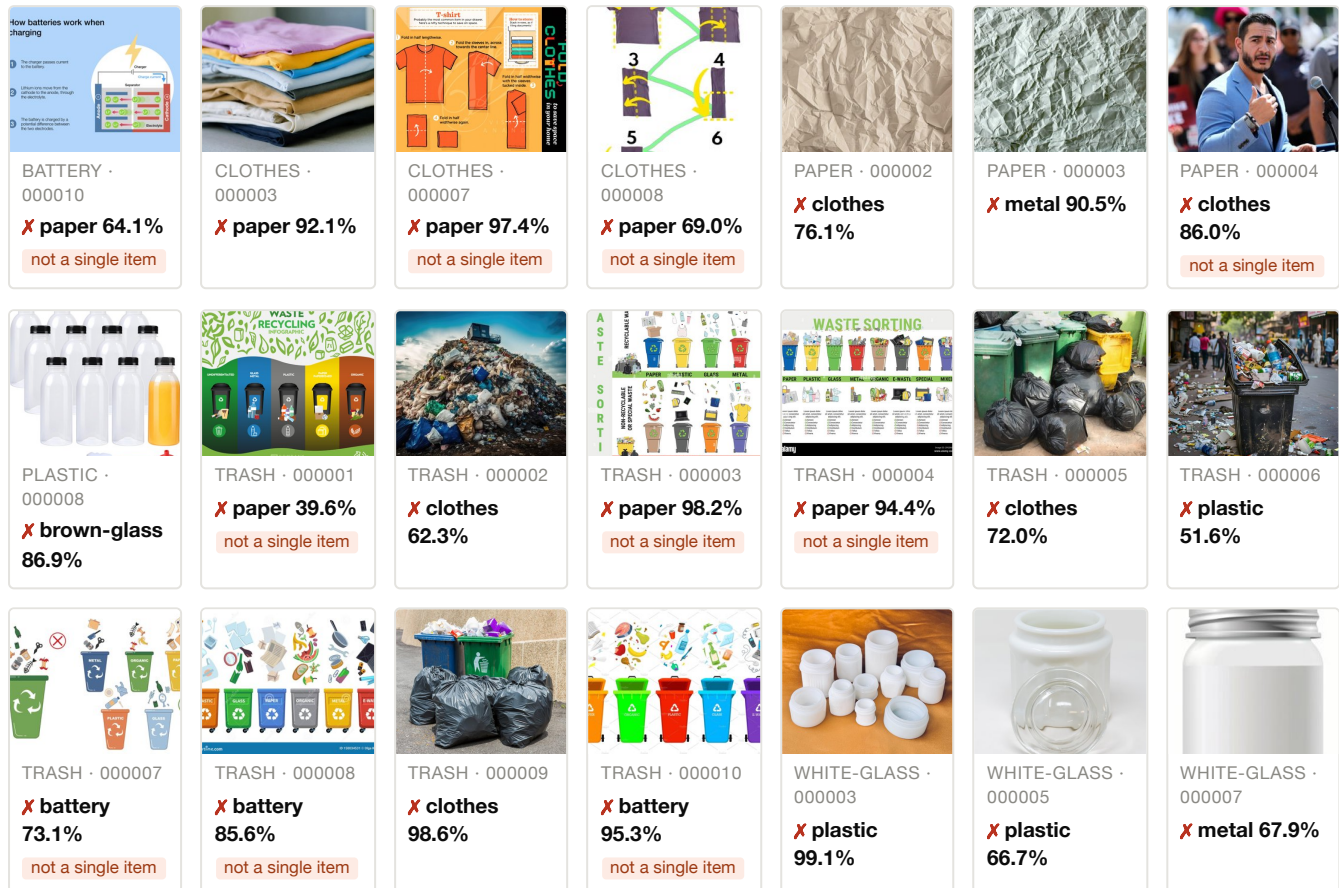
First image of every class

The `000001.jpg` image of each class, with its top prediction and the next two. ✓ correct · ✗ wrong.

 BATTERY · 000001 ✓ battery 99.1% brown-glass 0.9% · cardboard <0.1%	 BIOLOGICAL · 000001 ✓ biological 99.0% paper 0.8% · clothes 0.2%	 BROWN-GLASS · 000001 ✓ brown-glass >99.9% white-glass <0.1% · metal <0.1%	 CARDBOARD · 000001 ✓ cardboard >99.9% paper <0.1% · brown-glass <0.1%	 CLOTHES · 000001 ✓ clothes >99.9% paper <0.1% · white-glass <0.1%	 GREEN-GLASS · 000001 ✓ green-glass >99.9% brown-glass <0.1% · white-glass <0.1%
 METAL · 000001 ✓ metal >99.9% white-glass <0.1% · plastic <0.1%	 PAPER · 000001 ✓ paper 89.1% white-glass 4.0% · clothes 3.3%	 PLASTIC · 000001 ✓ plastic 99.9% brown-glass <0.1% · white-glass <0.1%	 SHOES · 000001 ✓ shoes 99.4% clothes 0.6% · metal <0.1%	 TRASH · 000001 ✗ paper 39.6% battery 34.9% · cardboard 25.4%	 WHITE-GLASS · 000001 ✓ white-glass 89.4% brown-glass 4.7% · plastic 3.4%

9 Sample predictions: where the model goes wrong

All 21 misclassified sample images. Each card shows the true class and file number, then the model's wrong answer and its confidence. Cards tagged **not a single item** are images that, on inspection, don't show one waste item.



Patterns in the mistakes

- Many errors are bad samples, not bad predictions.** 10 of the 21 misclassified images don't show a single waste item. They are recycling-bin infographics, clothes-folding guides, a battery-charging diagram and a photo of a person saved as "paper". The web scraper collected them without any checking. Treat the 82.8% as a lower bound for photos of real items.
- Scenes instead of objects.** The four genuine trash photos show landfills and bin bags. The model can name only one object per image, so it latches onto what stands out: bin bags read as clothes, for example `trash/000009` as clothes at 98.6%.
- Flat, soft textures get confused.** Paper is the most common wrong answer (7 times) and clothes the second (5). A stack of folded shirts is called paper at 92.1%, and crumpled paper is called clothes or metal.
- Clear glass looks like plastic.** Two white-glass jars are called plastic, and a jar with a metal lid is called metal. This matches the plastic / white-glass confusion on the test set (Section 6).
- High confidence doesn't mean correct.** Several wrong answers are above 90% confident, so a "not sure" threshold alone wouldn't catch them. Cleaner, in-the-wild training data would (Section 10).

10 Limitations, future work & conclusion

Limitations

- **Domain shift.** Training images mostly show one object on a plain background. Real bins are cluttered and poorly lit (Section 7).
- **Confusable materials.** Plastic, white glass and metal are the weakest classes even on clean data.
- **Single-label output.** One image gets one class, so a mixed pile can't be described.
- **Ambiguous class definitions.** "Trash" and "paper" mean different things in different contexts, as the web-image results showed.
- **Small real-world sample.** The 120 scraped images are unverified, so the real-world numbers are indicative only.

Future work

IDEA	EXPECTED BENEFIT
Train on in-the-wild images (hard-negative mining, cluttered scenes)	Close the 97% → 77% domain gap
Object detection (for example YOLO) for mixed waste	Classify every item in a scene, not just one
EfficientNet or ConvNeXt backbones, LR scheduling, early stopping	Higher accuracy, especially on plastic and white glass
Grad-CAM heatmaps in the app	Show users what the model looked at, which builds trust
Top-3 view and a "not sure" state below a confidence threshold	Safer behaviour on ambiguous inputs such as the low-confidence trash scenes (Section 9)
Export MobileNetV2 to ONNX or TorchScript	On-device inference for smart bins

Conclusion

RecycleVision shows that **transfer learning turns a modest dataset into a highly accurate waste classifier**. A fine-tuned ResNet50 reaches 97.4% accuracy and 0.968 macro F1 across 12 imbalanced classes, well above the 78% of a CNN trained from scratch. It is deployed as a public web app that anyone can try. The real-world test gives a candid view of where it stands: reliable on clear single items, with a clear path (in-the-wild training data and object detection) toward the cluttered conditions of real smart-bin and sorting-line deployments.

Tech stack

Language	Python 3.11	Visualisation	Matplotlib
Deep learning	PyTorch, torchvision	App & hosting	Streamlit, Streamlit Community Cloud
Data & metrics	NumPy, pandas, scikit-learn	Data collection	icrawler (Bing image search)
Hardware	Apple Silicon GPU (MPS) for training; CPU for inference		

A Appendix: reproducing the project

Repository layout

01_eda.ipynb	Exploratory data analysis
02_baseline_cnn.ipynb	CNN trained from scratch
03_transfer_learning.ipynb	MobileNetV2 + ResNet50 (freeze → fine-tune)
04_evaluation.ipynb	Test metrics, confusion matrices, comparison
app.py	Streamlit app (ResNet50)
download_real_world_dataset.py	Bing scraper for real-world images
helper_functions/data_loading/data_utils.py	get_dataloaders()
models/	BaselineCNN class + trained weights (.pt)
reports/figures, reports/metrics	Plots and JSON results
sample_images/, real_world_test/	122 web images (10–12 per class)

Run the app locally

```
git clone https://github.com/mukundnaramesh2605/RecycleVision--Garbage-Image-Classification-Using-Deep-Learning.git
cd RecycleVision--Garbage-Image-Classification-Using-Deep-Learning
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
streamlit run app.py # then open http://localhost:8501
```

Retrain from scratch

1. Download the [Kaggle dataset](#) into `dataset/garbage_classification/`.
2. Install the notebook dependencies: `pip install torch torchvision scikit-learn matplotlib pandas jupyter`.
3. Run the notebooks in order (01 → 04) from the project root. The split is fixed with `seed=42`.

Confusion matrices: baseline and MobileNetV2

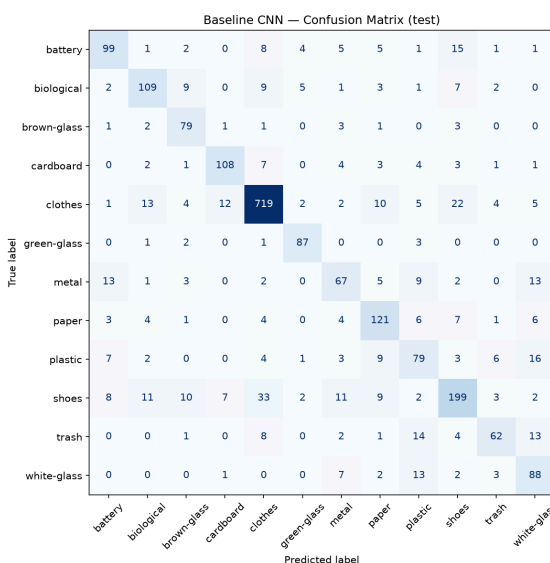


Figure 8. Baseline CNN (test accuracy 78.1%).

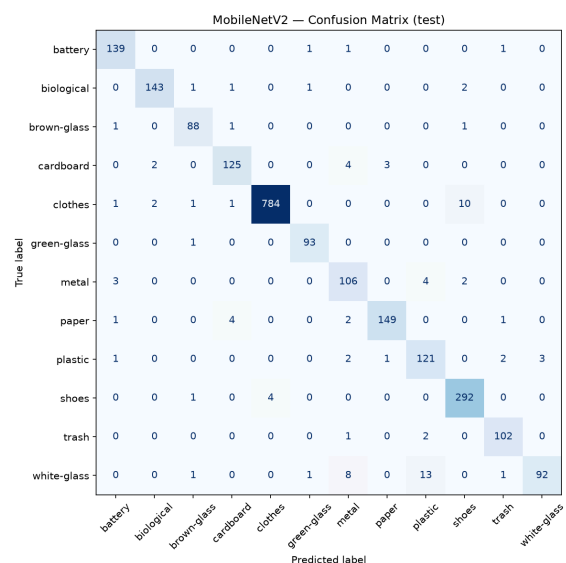


Figure 9. MobileNetV2 (test accuracy 96.0%).